



Hemostasis, Coagulation

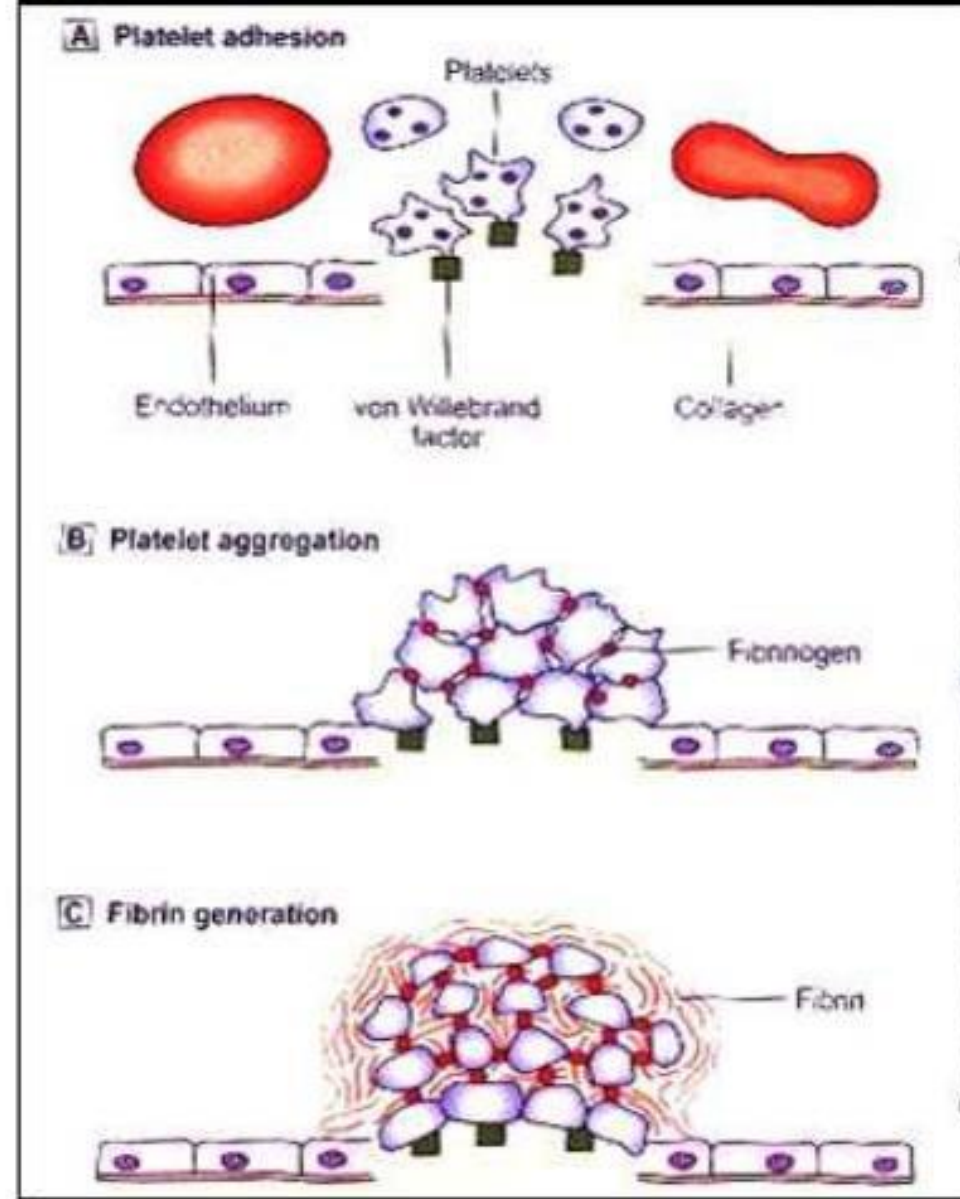
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Hemostasis

- Hemostasis is an amazing and complex process that keeps blood fluid in circulation and then, when an injury occurs, produces a clot to stop the bleeding, keeps the clot confined to the site of injury, and finally, dissolves the clot as the wound heals.
- Any disturbance or imbalance in the hemostatic system results in thrombosis (clotting) or hemorrhage (bleeding) which may be life-threatening.
- Understanding the major hemostasis systems—blood vessels, platelets, and plasma proteins—is essential to interpreting laboratory test results and preventing, predicting, diagnosing, and managing hemostatic disease.

Haemostasis:

- ☐ Vasoconstriction
- ☐ Platelet activation
- ☐ Haemostatic plug
- ☐ Coagulation
- ☐ Stable clot formation
- ☐ Clot dissolution



➤ OVERVIEW OF HEMOSTASIS: Hemostasis involves the interaction of vasoconstriction, platelet adhesion and aggregation, and coagulation enzyme activation to stop bleeding.

Hemostasis

Hemostasis is traditionally divided into two systems: **primary hemostasis** and **secondary hemostasis**.

I. Primary Hemostasis

- Involves platelets and vWF and results in forming a **platelet plug (primary hemostatic plug)**.
- If the endothelial injury is small, this may be adequate to stop bleeding.
- However, if the injury is greater, participation by the coagulation cascade is required.

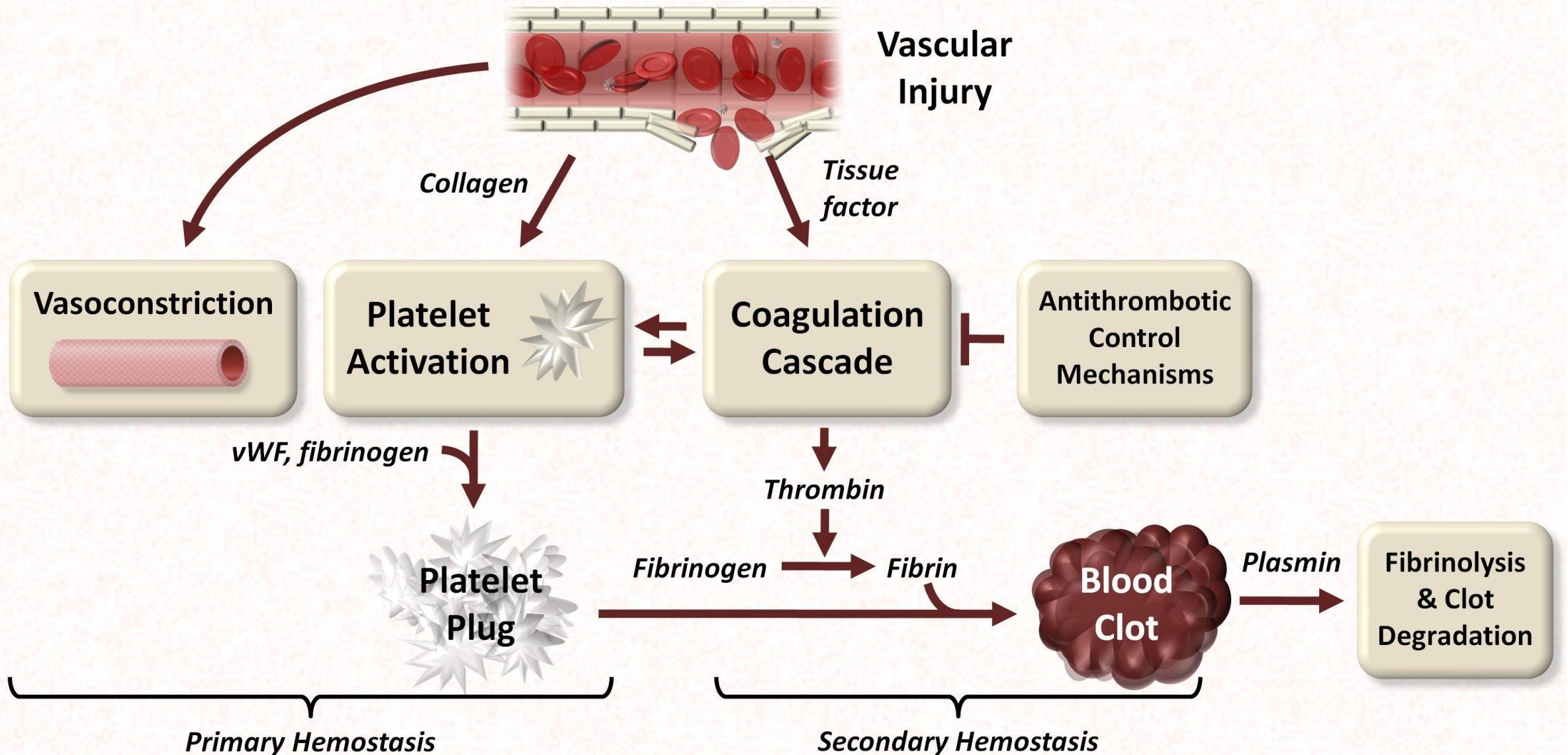
- **Primary hemostasis** refers to the role of blood vessels and platelets in response to a vascular injury.
- Blood vessels contract to seal the wound or reduce the blood flow (vasoconstriction).
- Platelets become activated, adhere to the site of injury, secrete the contents of their granules, and aggregate with other platelets to form a platelet plug.
- Vasoconstriction and platelet plug formation comprise the initial, rapid, short-lived response to vessel damage, but to control major bleeding in the long term, the plug must be reinforced by fibrin.

II. Secondary Hemostasis

- Involves the coagulation cascade proteins, which ultimately results in the conversion of fibrinogen to fibrin; fibrin polymerizes to form a **clot (fibrin clot or secondary hemostatic plug)**.
- The fibrin clot is cross-linked and stabilized by factor XIIIa.
- **Secondary hemostasis** describes the activation of a series of coagulation proteins in the plasma to form a fibrin clot.

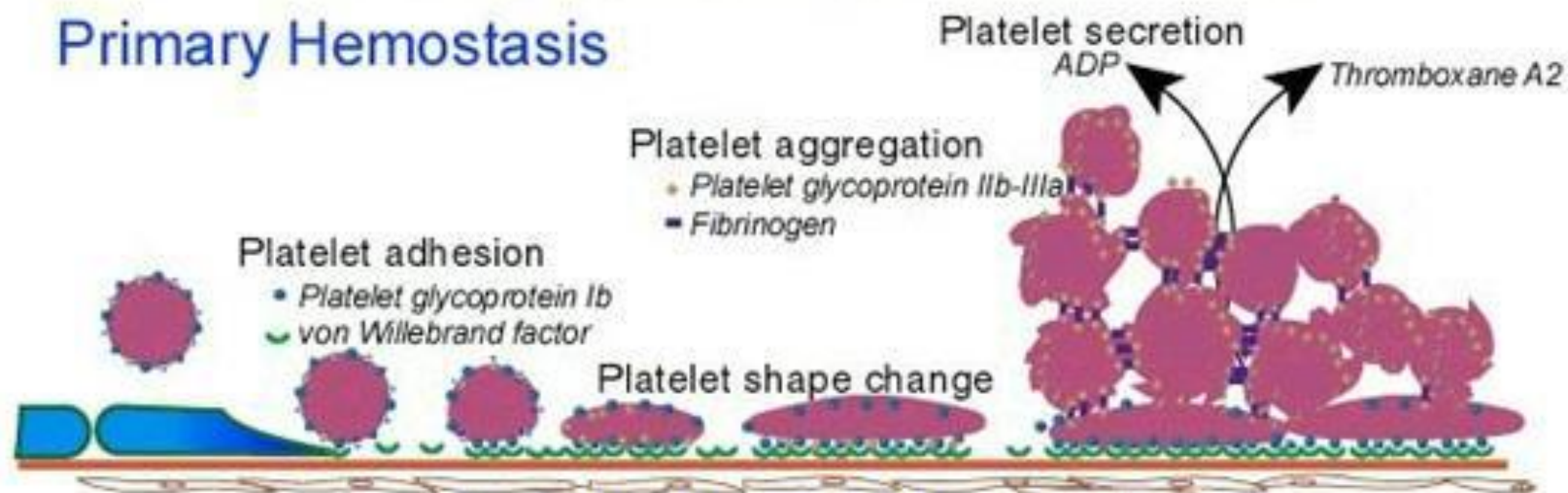
- These proteins circulate as inactive zymogens (proenzymes) that become activated during the process of coagulation and, in turn, form complexes that activate other zymogens to ultimately generate thrombin, an enzyme that converts fibrinogen to a localized fibrin clot.
- The final event of hemostasis is fibrinolysis, the gradual digestion, and removal of the fibrin clot as healing occurs.
- Although the vascular intima and platelets are associated with primary hemostasis, and coagulation and fibrinolysis are associated with secondary hemostasis, all systems interact in early- and late-hemostatic events.

Major Components of Hemostasis



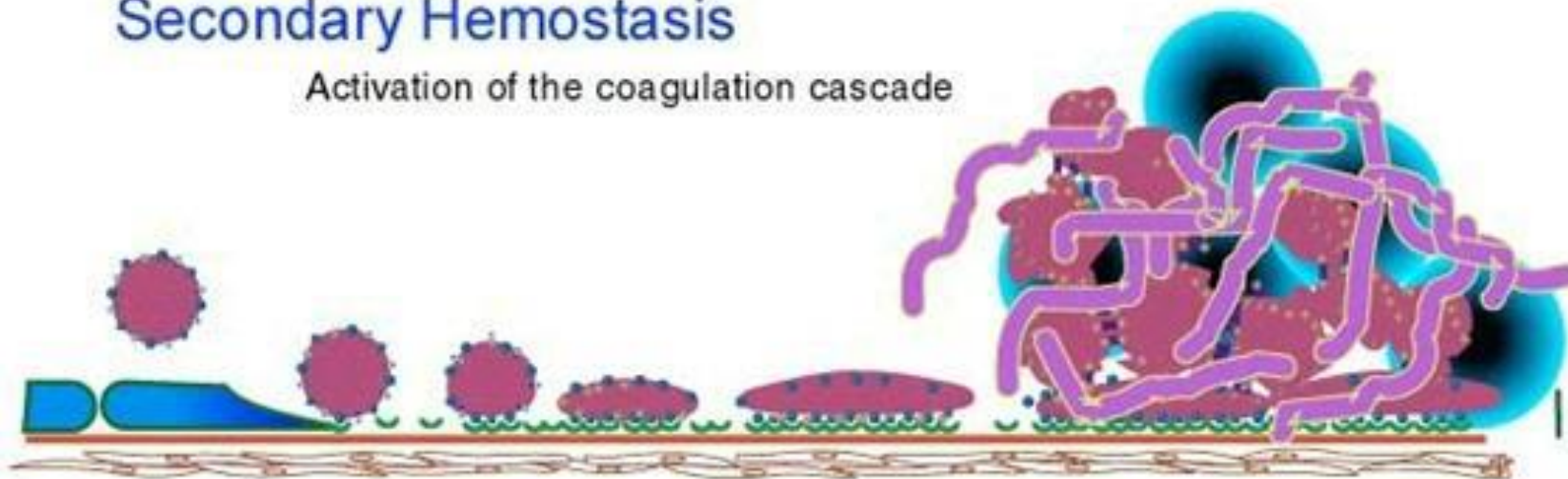
Formation of a Hemostatic Plug

Primary Hemostasis

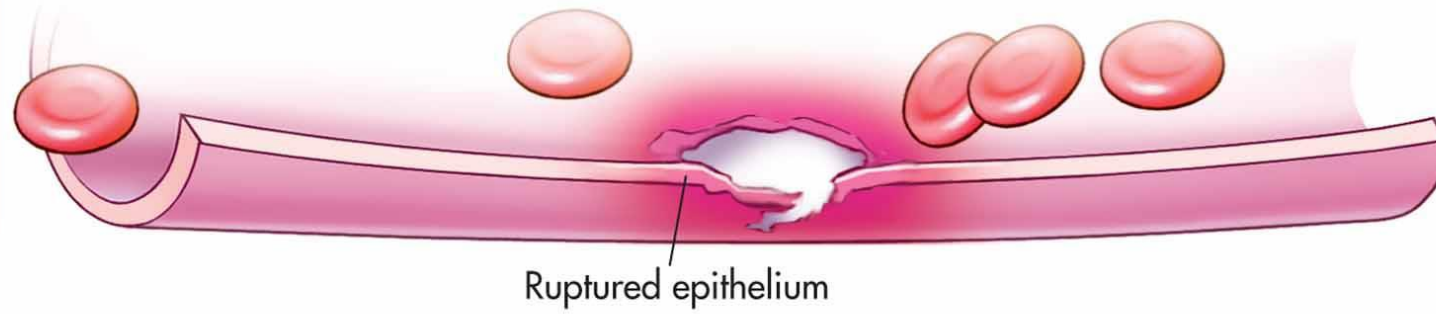


Secondary Hemostasis

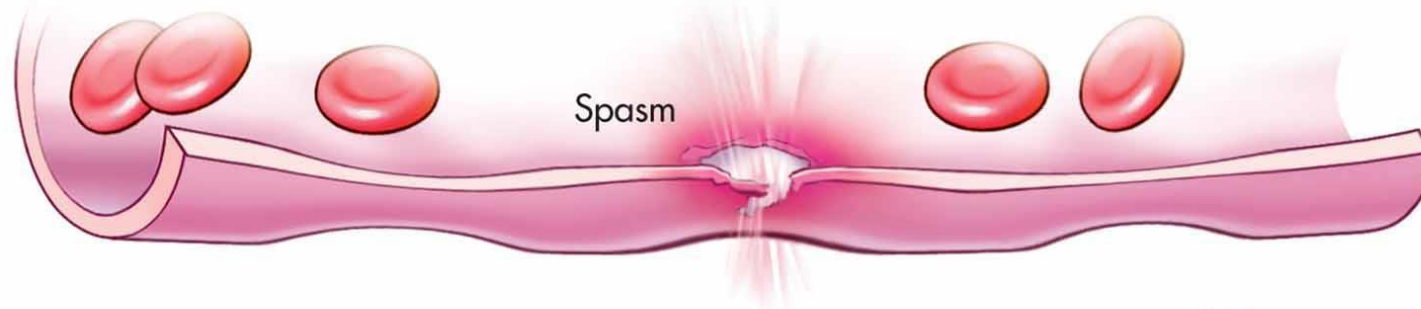
Activation of the coagulation cascade



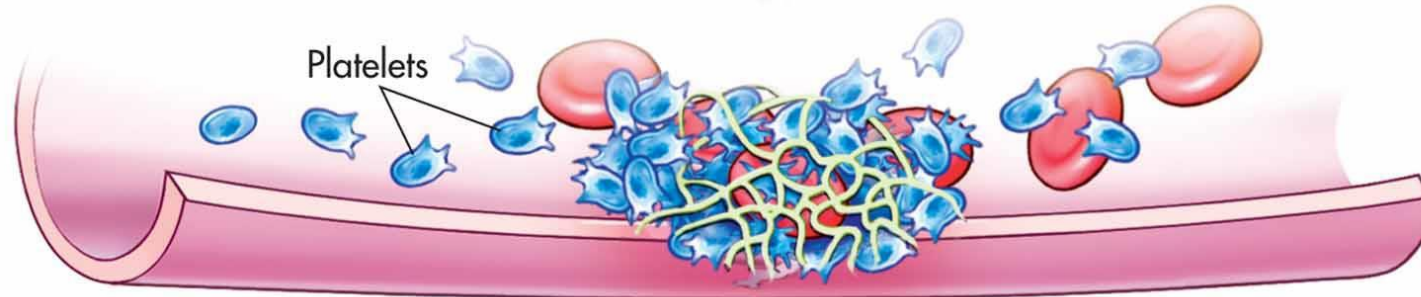
Vessel injury



Vessel spasm

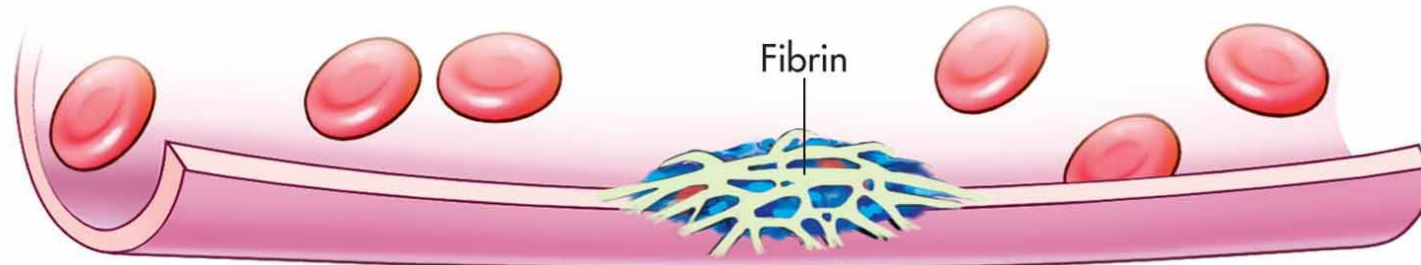


Platelets adhere to injury site and aggregate to form plug

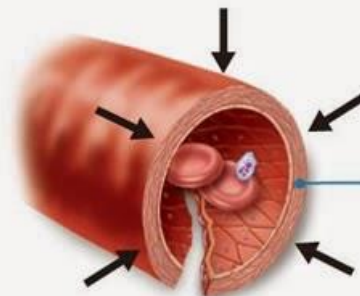


Platelets plug

Formation of insoluble fibrin strands and coagulation

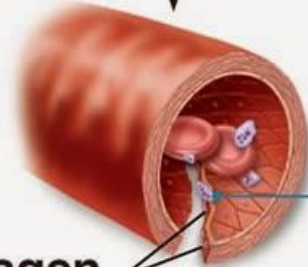


Fibrin clot



Step ① Vascular spasm

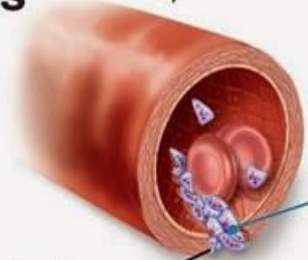
- Smooth muscle contracts, causing vasoconstriction.



Step ② Platelet plug formation

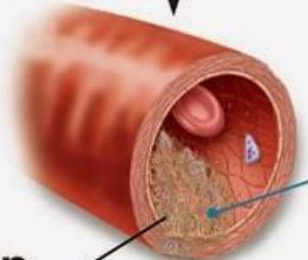
- Injury to lining of vessel exposes collagen fibers; platelets adhere.

Collagen fibers



- Platelets release chemicals that make nearby platelets sticky; platelet plug forms.

Platelets



Step ③ Coagulation

- Fibrin forms a mesh that traps red blood cells and platelets, forming the clot.

Fibrin

Hemostasis

There are two fundamental properties of the hemostatic system: exquisite regulation and control and surface dependence.

1- Regulation and Control Mechanisms

- Each system is balanced by an opposing or inhibiting system
- There are multiple negative feedback loops
- A system and its opposing system are often initiated simultaneously (A system often initiates its inhibiting system).

2- Surface Dependence

- Hemostasis is designed to occur on a phospholipid surface. This serves to limit clotting to the sites of tissue injury and prevents clotting from occurring freely in the circulation. Platelet phospholipid serves as the primary surface for coagulation.
- **Calcium dependence** is another feature of the coagulation systems. Calcium serves as the link between the platelet phospholipid and the coagulation proteins and is also critical in platelet activation and aggregation.
- **We take advantage of this by using calcium chelators as anticoagulants in our sample tubes.**

Components Of The Hemostatic System

Components of the global hemostatic system include:

1. Platelets
2. von Willebrand factor (**vWF**)
3. Tissue factor: has a critical role in initiating the coagulation cascade
4. Clotting or coagulation factors (proteins of the coagulation cascade)
5. The fibrinolytic system (plasminogen/plasmin, tissue plasminogen activator)
6. Anticoagulant or inhibitor proteins (anti-thrombin, protein C, protein S)
7. Blood vessels

Other less obvious factors are also important in hemostasis:

- **Vasoconstriction**; this decreases blood loss and allows a buildup of activated clotting factors in the injured area.
- **Blood flow**; which helps dilute the activated clotting factors and washes them away from the injured area. This helps to limit clotting to the appropriate area.
- **Endothelial cells** of the blood vessels.

Why coagulation doesn't occur in the normal state?

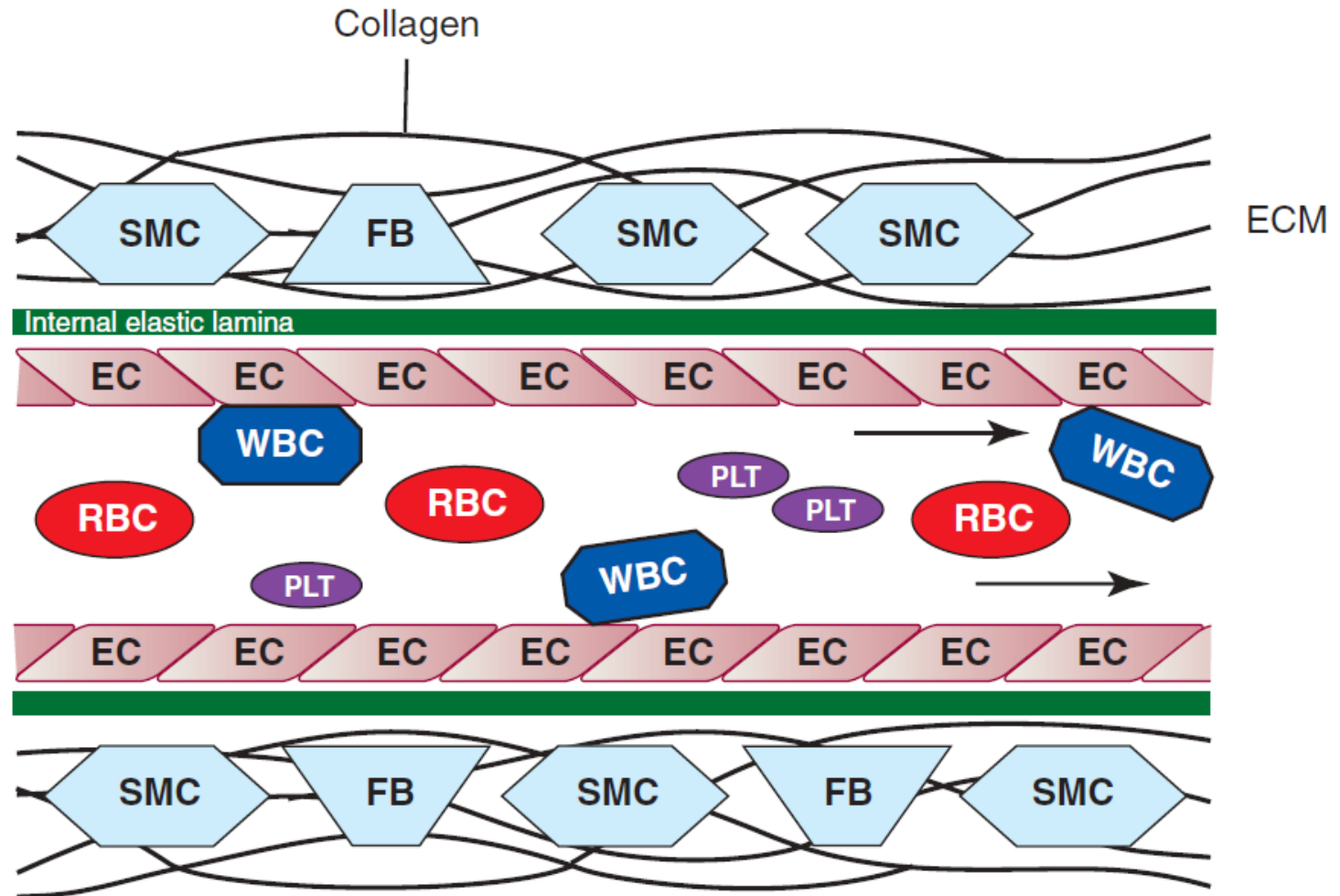
- **Endothelial cell surface molecules:** Endothelial cells express several molecules on their surface membranes that are important in regulation of coagulation. Examples are **heparan sulfate** and **thrombomodulin**, which activate anticoagulant systems (anti-thrombin and the protein C-protein S system, respectively).
- **Endothelial cell metabolic products:** Endothelial cells produce a variety of metabolic products that are critical in the prevention of thrombosis (prostacyclin).

- Tissue plasminogen activator (**TPA**), the primary initiator of the fibrinolytic system
- Tissue factor pathway inhibitor (**TFPI**), which inhibits coagulation via the TF-VIIa-Xa complex.
- Prostacyclin (**PGI₂**), a potent vasodilator and platelet antagonist.
- Endothelial cells also produce nitric oxide, which is a potent vasodilator and platelet antagonist.

- All coagulation factors are found in an inactive form (zymogen), and should be activated to coagulate the blood.
- The smooth surface of the endothelium surface doesn't allow the platelets to adhere and aggregate.

Normal ECM suppresses hemostasis:

- Prostacyclin
- Heparan sulfate
- Tissue factor pathway inhibitor
- Nitric oxide
- Thrombomodulin



Coagulation Cascade

- The classic concept of the coagulation cascade featured two separate and independent pathways: the **extrinsic pathway** and the **intrinsic pathway**.
- The two pathways came together at the activation stage of factor X to Xa, and hence the pathway from factor X down to fibrin was called the **common pathway**.
- One coagulation factor cleaves and activates the next factor along the line and so on.

- Since each active enzyme can activate many molecules of the subsequent factor, there is a geometric increase in the number of molecules activated.
- Like a small snowball starting at the top of a hill, the end result of the coagulation cascade is an avalanche of activated clotting factors.
- Coagulation means that the blood is **converted from the liquid or fluid state to the gel form.**
- Coagulation cascade needs calcium and phospholipid.

Coagulation Cascade: Classic Concept

Extrinsic Pathway

Tissue Factor

VII → VIIa

Intrinsic Pathway

Negatively Charged Surface

XII → XIIa

XI → XIa

IX → IXa

VIIIa, Phospholipid, Calcium

Common Pathway

X → Xa

Va, Phospholipid, Calcium

Prothrombin (II) → Thrombin (IIa)

XIII → XIIIa

Fibrinogen → Fibrin (Soluble) → Insoluble Fibrin

Extrinsic Pathway

- The extrinsic pathway starts with exposure of **tissue factor (TF)** from subendothelium to blood and the subsequent activation of factors VII, X, V, II and I to form the fibrin clot.
- Tissue factor is a transmembrane protein that is highly expressed in the brain, kidney, and other tissues.
- TF is not normally present on endothelial surfaces or blood cells.
- It is named extrinsic pathway, because the TF is not present in the plasma like other coagulation factors.

- The key initiating step is the exposure of TF to the circulation and reaction of TF with factor VIIa.
- Exposed tissue factor reacts with trace amounts of factor VIIa, which are normally present in the circulation.
- The TF-factor VIIa complex can enzymatically activate factor X to Xa, factor IX to IXa, and factor XI to XIa.
- The TF-factor VIIa complex activates factor X to Xa, starting the common pathway.

Intrinsic Pathway

- **The intrinsic pathway** refers to the path of the coagulation cascade in which **Prekallikrein (PK), high-molecular-weight kininogen (HMWK), and factors XII, XI, X, IX, VIII, V, II, and I** are involved in the formation of a fibrin clot.
- The intrinsic pathway (also called the **contact activation pathway**) is called that, because the initiator factor for it is present in the plasma (factor XII).

➤ **Initiation** of the intrinsic system coagulation cascade is by the **activation of factor XII (Hageman factor)** coming in contact with a negatively charged surface and being activated to XIIa.

1. Vascular damage exposes **negatively charged** subendothelial tissue.
2. The inactive **zymogen** form of factor XII is attracted to the negatively charged endothelial surface of the damaged blood vessel.
3. The negative polarity activates factor XII to XIIa.
4. **HMWK** and **PK** are required.

➤ Factor XIIa then activates XI to XIa, XIa activates IX to IXa, IXa activates X to Xa in the presence of factor VIIIa, calcium, and phospholipid (Tenase complex), starting the common pathway.

